Prodigy Anesthesia Quick Review

Anticoagulants

What is the difference between an anticoagulant and an antithrombotic?	Anticoagulants such as heparin and enoxaparin delay or prevent the clotting of blood by interfering with the coagulant system. Antithrombotic agents such as aspirin typically interfere with with thrombus formation by interfering with the adherent or aggregation properties of platelets. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 505.*
What are the principal uses of heparin?	The treatment and prevention of deep vein thrombosis, pulmonary embolus, mural thrombosis following myocardial infarction, in the treatment of unstable angina and infarction, and the prevention of thrombosis due to exposure of the blood to extracorporeal circulation devices such as dialysis and cardiopulmonary bypass. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 506.*
How does heparin administration affect the pharmacokinetics of diazepam and propranolol?	Heparin displaces alkaline drugs from their protein binding sites and can increase the circulating concentrations and clinical effects of these drugs. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 509.*
Following the administration of heparin, a patient develops fever, tachycardia, and urticaria. What do you believe is occurring?	Because heparin is derived from animal sources, some patients may be allergic to it. It should be used cautiously in patients with known allergies to porcine or bovine derivatives. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 509.*
What is HITT syndrome?	Heparin-induced thrombocytopenia and thrombosis syndrome, which is a potentially life-threatening disorder that occurs in about 5% of patients receiving heparin. The patient develops severe thrombocytopenia, becomes resistant to the effects of heparin, and may develop thombosis despite the low platelet count. It usually develops after 6-10 days of heparin therapy. It is due to the development of platelet-associated immunoglobulin (IgG) antibodies. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 508.*
What is the onset of action of IV heparin? Subcutaneous?	The onset is immediate with intravenous heparin and 1-2 hours when administered SQ. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 506.*

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What ACT value is considered indicative of adequate heparinization for cardipulmonary bypass?	An ACT >300 seconds is considered adequate, 180-300 seconds is considered questionable, and less than 180 seconds is considered inadequate. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 506.*
What two laboratory values are used to measure the efficacy of heparin therapy?	The activated plasma thromboplastin time (APTT) and the activated coagulation time (ACT). The therapeutic range for the APTT is considered to be 1.5-2.5 times the normal value of 30-35 seconds. The ACT is used more commonly during surgical procedures because of its ease of use. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 506.*
What antihypertensive agent can increase the required dose of heparin?	Nitroglycerin *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 506.*
What factors can prolong the half-life of heparin?	Hepatic dysfunction, renal dysfunction, and decreases in body temeprature below 37 degrees Celsius *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 505.*
What is the elimination half-time of 100 U/kg of intravenous heparin?	About 56 minutes *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 505.*
Why must heparin be administered by injection?	Heparin has a high molecular weight (ranging between 3,000 and 30,000 daltons) and is poorly lipid soluble. Because of this, it is poorly absorbed from the gastrointestinal tract. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 505.*
Where is heparin located naturally within the body?	It is located in basophils, mast cells, and the liver (from which it derives its name). *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 505.*

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How does heparin produce its anticoagulant effect?	It binds to antithrombin, a natural anticoagulant, and enhances the ability of antithrombin to inactivate thrombin and activated factors X, XII, IX, and XI by over 1000 times. Heparin also inhibits platelet function. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 505.*
What patients would have a higher risk for having an allergic reaction to protamine?	Patients with a history of exposure to protamine zinc insulin and men who have undergone vasectomies have an increased risk for allergic reaction to protamine, especially if they have a history of allergic reaction to the insulin. Also, because protamine is derived from salmon milt, patients with a history of allergy to fish products have an increased risk for allergic reaction to protamine. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1489.*
What are the contraindications to heparin administration?	Heparin should not be administered to patients with a known bleeding diathesis or those presenting for intraocular or intracranial surgery. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 508.*
How long after the administration of a low molecular weight heparin should you wait prior to surgery?	Because they have a longer half-life than heparin and a risk of bleeding that is nearly that of heparin, surgery should be delayed until 12 hours after the last dose if at all possible. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 511.*
How does protamine antagonize the effects of heparin?	The positively charged and alkaline protamine binds with the negatively-charged and acidic heparin molecule to form a highly stable compound that has no anticoagulant properties. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 510.*
What is the estimated dose of protamine required to reverse the effects of heparin?	1 mg of protamine for every 100 units of heparin predicted to still be circulating. (Remember that heparin has a half-life of about an hour) *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 510.*
What are the potential side effects of rapid administration of heparin?	Histamine release with associated hypotension, flushing, tachycardia. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 510.*

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Protamine can potentially stimulate the release of thromboxane and serotonin. What are the side effects seen if this occurs?	Pulmonary vasoconstriction, bronchoconstriction, pulmonary hypertension, and arterial hypoxemia. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 510.*
How do heparin and the low molecular weight heparins (enoxeparin and dalteparin) compare with respect to their protein binding ability? What pharmacokinetic effect does this have?	Heparin is highly protein-bound and binds to numerous types of protein, the concentrations of which can vary widely among patients. This results in a wide variance in the dose-response relationship between patients. The low molecular weight heparins do not bind as avidly to plasma proteins. As a result, they have a much more stable dose-response profile. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 511.*
How does the half-life of low molecular weight heparin compare to that of heparin?	Low molecular weight heparins have a half life of 4-5 hours compared to about 1 hour for heparin. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 511.*
A patient on warfarin is undergoing a preoperative evaluation for major surgery. How long prior to surgery should they discontinue their warfarin?	Warfarin should be discontinued 1-3 days prior to major surgery so that the PT can return to within 20% of its normal value. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 513.*
Oral anticoagulants are all derivatives of what chemical?	Coumarin *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 511.*
A patient on warfarin has an INR of 3.0 and is presenting for emergency surgery following a motor vehicle accident. He last took his medicine 8 hours ago. How can you reduce the incidence of bleeding in this patient?	The administration of vitamin K 1-2 mg IV can reverse the anticoagulant effects of warfarin within about 4 hours. For even faster reversal, or to prevent bleeding in higher risk surgeries such as craniotomy, you can administer recombinant factor VIIa or 1-2 units of fresh frozen plasma. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 513.*

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What laboratory test is used to guide therapy with warfarin and other oral antocoagulants?	The prothrombin time, which reflects the activity of three of the four vitamin K factors is used to guide oral anticoagulant therapy. Because commercial reagents vary so much, the International Normalized Ratio (INR) is used as a standard to correct for these variances. The target INR for most clinical uses of oral anticoagulants should be 2.0 to 3.0. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 513.*
Does warfarin cross the placenta? Does it cross into breast milk?	Warfarin does cross the placenta and can have severe effects on a fetus because of an already immature ability to form clotting factors. Because it is 97% protein bound, it does not cross over into breast milk. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 513.*
How long after the ingestion of warfarin does it reach peak plasma concentrations? How long til it begins to affect coagulation? Why is there a difference between the two?	Warfarin reaches a peak plasma concentration about 1 hour after ingestion. The peak effects, however, are not reached for another 24-72 hours which reflects the half-life of the already formed vitamin K-dependent clotting factors. Once these factors are used up, the deficit of vitamin K will have an inhibitory effect on coagulation. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 512-513.*
How does warfarin affect platelet activity?	Warfarin only affects vitamin K and does not affect platelet activity. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 512.*
How does warfarin inhibit coagulation?	It inhibits an enzyme known as vitamin K epoxide reductase. By blocking this enzyme, it prevents the conversion of vitamin K epoxide into vitamin K. Clotting factors II, VII, IX, and X all depend upon vitamin K. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 512.*
How do phosphodiesterase inhibitors exert their anticoagulant effects?	Phosphodiesterase inhibitors increase the level of cyclic AMP which is an inhibitor of platelet aggregation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 437.*
What is the prinicipal indication for thrombolytic therapy?	It is used primarily in the treatment of acute coronary syndrome to dissolve the offending thrombus. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 515.*

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What is a common abnormal laboratory finding in patients taking ximelagatran?	It can cause elevated transaminase enzyme levels *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 516.*
How long prior to surgery should ticlopidine be discontinued?	7-10 days *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 342.*
How long prior to surgery should clopidogrel be discontinued?	7 days *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 517.*
How does the mechanism of action of aspirin differ from that of clopidogrel and ticlidopine?	They are all adenosine diphosphate inhibitors, but aspirin inhibits the release of adenosine diphosphate by blocking Cox-1 and Cox-2 receptors and clopidogrel and ticlopidine block the ADP receptors on the surfaces of the platelets which inhibits platelet activation. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 517.*
How do glycoprotein IIb/IIIa antagonists inhibit clotting?	Glycoprotein IIb/IIIa antagonists such as abciximab prevent fibrinogen from binding to the glycoprotein IIb/IIIa receptors on platelets. By doing so, they prevent platelets from aggregating. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 517.*
What is the advantage of ximelagatran over heparin?	Ximelagatran is not associated with an increased risk of spontaneous bleeding. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 516.*
What is the indication for ximelagatran?	It is used to prevent perioperative deep vein thrombosis. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 516.*

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What are the advantages of hirudin over heparin?	Hirudin is capable of inhibiting thrombin that is both free in the circulatory system and bound within a thrombus, making it effective for both the treatment and prevention of thromboembolic disease. Unlike heparin, it is not associated with an increased risk of spontaneous bleeding and is not associated with immune-related thrombocytopenia. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 516.*
What is the primary risk associated with the administration of a thrombolytic and in what patients should it be avoided?	The primary risk is spontaneous bleeding. Thrombolytics are contraindicated in patients who have recently experienced trauma, had surgery or invasive procedures, have a history of gastrointestinal bleeding or have had a recent hemorrhagic stroke. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 515.*
How do thrombolytic drugs work?	They convert plasminogen into the fibrinolytic enzyme plasmin. *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 514.*
What class of drugs are hirudin, ximelagatran, and argatroban?	They are direct thrombin inhibitors (antithrombotic drugs) *Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 516.*

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